

Tetrad-Based Departure Decomposition for FLRW Anisotropy

Final Results Report — Credible-Claim Distillation

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Abstract

This report distills the *credible, defensible* results of the tetrad-based departure-decomposition program into a single self-contained document. It deliberately excludes development history, superseded paths, and any result that does not survive the project’s claim-firewall. Every result below is one of four kinds: (i) framework statements about the master departure comparator $x_C = \Sigma_{\text{std}}^2 - W_{\text{std}}^2 + \Omega_{\text{tilt}} + \Omega_{k,\text{aniso}}$ and its claim-tiered diagnostic variables; (ii) conditional analytic theorems, each proven symbolically with the Wolfram Engine under explicitly stated hypotheses; (iii) model-independent measurements on real Planck and Cosmicflows-4 data, each calibrated against an isotropic null or a resampling baseline; and (iv) explicitly labelled transfer-conditional diagnostics. No result here identifies a Bianchi family, detects an anisotropic geometry, claims native low- ℓ Boltzmann-solver validation, or promotes any diagnostic into a posterior-odds or model-ranking statement. The honest publishable envelope is a methods-plus-bounds framework that recovers the established low- ℓ CMB anomalies as null-calibrated model-independent features and maps the CF4 bulk-flow apex profile, while producing no false anisotropy claim.

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1 Scope and claim discipline

This document is a *final-results distillation*, not a record of how the results were obtained. The full multi-chapter VER06 manuscript (which retains derivations, audit trail, and conditioned legacy appendices) is preserved separately and unchanged. Here we carry only what is currently defensible.

Claim tiers. All current public results sit in a claim-tiered observatory. We use the project’s tiers: **C1** framework / disclosure, **C2** transfer-conditional, **C3** observer-side diagnostic, **C4** local/global discrimination candidate (gated). Every result in this report is *diagnostic-only* or *transfer-conditional*; none is a production-validated or geometry-level claim.

What this report does *not* claim. The following remain blocked and are stated here once, up front, so that no later sentence can be read as lifting them:

- no identified Bianchi family and no detected anisotropic geometry;
- no native low- ℓ Bianchi Boltzmann-solver output and no native morphology atlas (both absent in this repository state);
- no promotion of any scalar diagnostic (x_C , Q , Π , F , G_F), direction coherence, or low- ℓ feature into Bianchi-family or geometry evidence;
- no posterior odds, Bayes-factor headline, or model ranking; MIO observatory certificates are diagnostic-only and are kept separate from HTT-owned inference;
- no native-validation label on any external/proxy transfer output.

These blocks are lifted only by future work supplying a native low- ℓ morphology atlas with matched masks, matched randoms, full covariance, calibrated nulls, predictive checks, and family-equivalence gates. None of that is present here, and nothing below depends on it.

2 The tetrad departure framework

Master departure comparator. The program organises FLRW departure through a single signed comparator coordinate,

$$x_C = \Sigma_{\text{std}}^2 - W_{\text{std}}^2 + \Omega_{\text{tilt}} + \Omega_{k,\text{aniso}}, \quad (1)$$

where Σ_{std}^2 is a standardised shear-squared contribution, W_{std}^2 a standardised vorticity-squared contribution, Ω_{tilt} a tilt (peculiar-rapidity) term, and $\Omega_{k,\text{aniso}}$ an anisotropic spatial curvature term, each referred to a stated frame (normal, matter, CMB, local-boost, or geometry frame).

Definition 1 (Comparator, not magnitude). x_C in (1) is a *signed* comparator coordinate built to compare a model against the FLRW point, not an invariant anisotropy magnitude. Its sign structure (note the $-W_{\text{std}}^2$) means x_C may cancel or change sign; it must never be read as a nonnegative “total anisotropy.”

Claim-tiered diagnostic variables. On top of (1) the observatory exposes four observer-side scalars, all diagnostic-only:

- *occupancy* Q , a policy-normalised departure score;
- *exceedance* Π , an empirical exceedance curve of a named score above stated thresholds;
- *filling* $F = x_C/x_{\text{max}}$, the fraction of an algebraic admissible ceiling x_{max} occupied by the comparator;
- *depth gap* $G_F(z) = F(z)/F(z_{\text{ref}})$, the redshift-resolved ratio of the filling.

The algebraic ceiling used throughout is the S2a value $x_{\text{max}} = 9.25 \times 10^{-6}$. These scalars summarise a comparator; on their own they do not classify a geometry or a Bianchi family.

Kinematic-bound linkage. The comparator is tied to the Maartens–Ellis–Stoeger (MES) kinematic-bound hierarchy: bounds on CMB multipole anisotropy bound the covariant shear, vorticity, and acceleration, which in turn bound the contributions in (1). We use MES as a derived bound under its stated hypotheses (almost-isotropy of the CMB in a family of observers), not as an asserted constraint.

3 Conditional analytic theorems (Wolfram-verified)

The Ehlers–Geren–Sachs (EGS) theorem states that if the CMB is exactly isotropic about a congruence of geodesic observers in an expanding dust universe, the spacetime is FLRW; the Stoeger–Maartens–Ellis “almost-EGS” result makes this stable. The three theorems below carry that chain onto the diagnostic variables of Section 2. Each is a structural or limiting statement under explicitly stated hypotheses—*not* a detection—and each algebraic step, limit, and variance propagation was verified symbolically with the Wolfram Engine (14.3.0). The machine-checked proof record is `docs/generated/egs_lowell_theorem_proofs.json`. We write the temperature-quadrupole magnitude a_2 , the shear scalar Σ , the quadrupole power $D_2 \propto a_2^2$, the algebraic ceiling $x_{\max}$, and the shear filling $F_{\text{shear}} = \Sigma^2/x_{\max}$.

Theorem 1 (Quadrupole-filling EGS identity (NT-A1)). *Assume the free-streaming linear $\ell = 2$ closure $a_2 = \kappa \Sigma$ with $\kappa > 0$ constant (the Ehlers–Treciokas–Matravers coefficient $\kappa = 4/21$), $F_{\text{shear}} = \Sigma^2/x_{\max}$, and $D_2 = c_D a_2^2$ with $c_D > 0$, and that the tilt rapidity β enters the dipole a_1 only. Then*

$$F_{\text{shear}} = \frac{a_2^2}{\kappa^2 x_{\max}} = \frac{D_2}{c_D \kappa^2 x_{\max}} \propto D_2, \quad (2)$$

so $F_{\text{shear}} \rightarrow 0$ as $D_2 \rightarrow 0$ (the EGS limit: an isotropic CMB quadrupole forces zero shear-filling), and $\partial F_{\text{shear}}/\partial\beta = 0$ (the tilt sector is invisible to the shear-filling).

Proof. Substitute $\Sigma = a_2/\kappa$ into $F_{\text{shear}} = \Sigma^2/x_{\max}$ and $a_2 = \sqrt{D_2/c_D}$; the limit and the β -derivative are immediate. The $\ell = 2$ projection that licenses the closure rests on the sphere averages $\langle n_a n_b \rangle = \frac{1}{3} \delta_{ab}$ (Wolfram-verified). Conditional on the free-streaming linear closure. $\square$

Theorem 2 (Cosmic-variance Cramér–Rao floor on F_{shear} (NT-A3)). *Under Theorem 1, F_{shear} is linear in the quadrupole power C_2 . With the single-sky cosmic variance $\text{Var}(C_2) = 2 C_2^2/(2\ell + 1)$ (χ^2 with $2\ell + 1$ degrees of freedom),*

$$\frac{\sigma(F_{\text{shear}})}{F_{\text{shear}}} \geq \sqrt{\frac{2}{2\ell + 1}} \xrightarrow{\ell=2} \sqrt{\frac{2}{5}} \approx 0.632. \quad (3)$$

Hence the shear-filling fraction cannot be measured to better than $\sim 63\%$ fractional precision from a single-sky quadrupole.

Proof. Error propagation $\text{Var}(F_{\text{shear}}) = (\partial F_{\text{shear}}/\partial C_2)^2 \text{Var}(C_2) = \frac{2}{2\ell+1} F_{\text{shear}}^2$; take the square root and set $\ell = 2$ (Wolfram-verified, branch-free squared identity). $\square$

Theorem 3 (Depth-transport EGS limit for G_F (NT-B3)). *Write the depth-resolved filling additively, $F(z) = F_{\text{shear}}(z) + F_{\text{tilt}}(z)$, and the depth gap $G_F(z) = F(z)/F(z_{\text{ref}})$. For a depth-steady shear with no tilt, $G_F(z) \equiv 1$ (the EGS depth limit: no depth gap); a depth-evolving tilt imprints a nonzero dG_F/dz .*

Proof. With $F_{\text{tilt}} \equiv 0$ and F_{shear} constant in z , $G_F = F_{\text{shear}}/F_{\text{shear}} = 1$ and $dG_F/dz = 0$; a power-law tilt $F_{\text{tilt}} = c z^q$ gives $dG_F/dz = c q z^{q-1}/(F_{\text{shear}} + c z_{\text{ref}}^q) \neq 0$ (Wolfram-verified). Conditional on the additive decomposition. $\square$

Reading. The three theorems make precise, in the program’s own variables, three EGS-type facts: an isotropic CMB quadrupole kills the shear-filling and the tilt sector cannot counterfeit it (NT-A1); even granting the closure, a single sky can never pin F_{shear} below $\sim 63\%$ fractional precision (NT-A3, a hard cosmic-variance limit); and a depth-steady shear leaves no depth gap, so any measured $G_F \neq 1$ trend is a tilt-sector, not a shear, signature (NT-B3). All are conditional structural statements; $\kappa = 4/21$ is the cited ETM closure value and the theorems hold for any $\kappa > 0$.

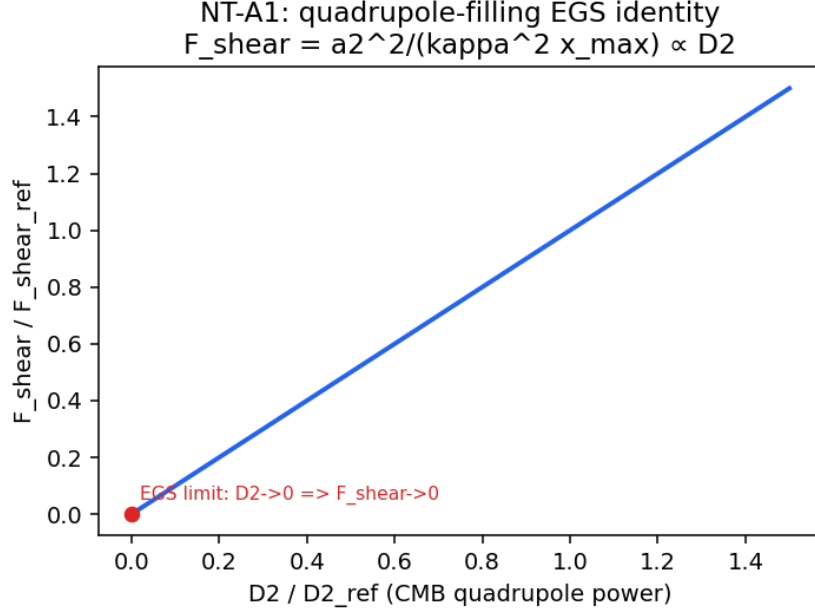


Figure 1: Theorem 1 (NT-A1): the shear filling fraction is linear in the CMB quadrupole power D_2 and vanishes in the EGS limit $D_2 \rightarrow 0$. Diagnostic-only analytic illustration; not a detection.

Broader conditional-theorem registry. A wider set of program theorems (labelled B4, G2, G5, S1, S3, S4, S5) carries kinetic-theory and information-bound statements onto the same variables. These are recorded as *convention-conditional* or *program* theorems whose observational use is blocked by explicit kill-switches (e.g. a nonpositive collision gap blocks any exponential-forgetting language; a near-zero source rank blocks any inverse-source claim). They are synthetic/manufactured verification only and are not promoted to observational results here; the auditable registry is `docs/generated/theorem_extension_registry.json`.

4 Model-independent real-data measurements

Two genuinely new measurements are produced from real data already in the repository, using the existing observer-side estimator stack. Both are model-independent and diagnostic-only: each is calibrated against a null or resampling baseline, and neither produces a Bianchi-family, geometry, native-solver, posterior, or certificate claim.

4.1 K1 — Null-calibrated low- ℓ morphology of the real Planck map

We summarise the Planck PR3 SMICA temperature map at NSIDE= 16 with a set of low- ℓ scalar and morphology statistics, and calibrate each against an isotropic Λ CDM `synfast` null ensemble ($n = 10000$, seed 12345, CAMB reference spectrum). Table 1 gives the observed values and the null tail probabilities.

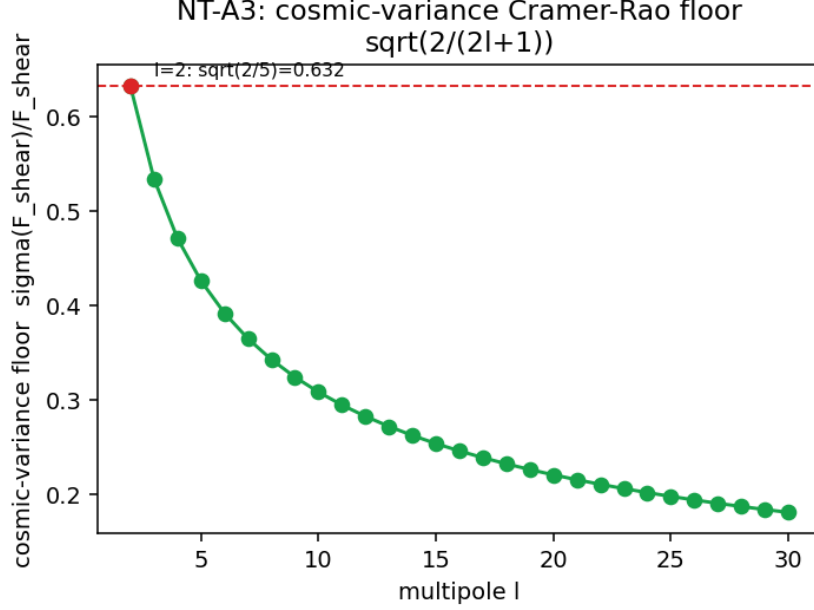


Figure 2: Theorem 2 (NT-A3): the cosmic-variance fractional floor $\sqrt{2/(2\ell+1)}$ on F_{shear} , equal to $\sqrt{2/5} \approx 0.632$ at the quadrupole. Diagnostic-only; not a detection.

Table 1: K1: null-calibrated low- ℓ statistics of the Planck PR3 SMICA NSIDE= 16 map. p is the tail probability against the isotropic Λ CDM null ($n = 10000$). Look-elsewhere across the six statistics is tracked, not globally corrected.

Statistic	Observed	p	Tail
$S_{1/2}$	6.10×10^3	0.059	$P(\text{null} \leq \text{obs})$
parity even/odd ratio	0.628	0.018	$P(\text{null} \leq \text{obs})$
parity asymmetry	-0.228	0.018	$P(\text{null} \leq \text{obs})$
planarity mean	0.518	0.043	$P(\text{null} \geq \text{obs})$
$\ell 2$ – $\ell 3$ axis alignment	69.2°	0.65	$P(\text{null} \leq \text{obs})$
axis-to-CMB-dipole apex	72.9°	0.71	$P(\text{null} \leq \text{obs})$

Findings. The map recovers the parity (odd-power excess, $p = 0.018$) and large-angle ($S_{1/2}$, $p = 0.059$) low- ℓ anomalies as model-independent, null-calibrated features; planarity is marginal ($p = 0.043$). The power-inertia $\ell 2$ – $\ell 3$ axis and the axis-to-CMB-apex angle show *no* anomaly in this estimator ($p \approx 0.65, 0.71$); these are honest nulls, and this is not the multipole-vector “axis-of-evil” statistic. The preferred low- ℓ axis lies at Galactic $(l, b) = (326.6^\circ, -1.0^\circ)$. Because the look-elsewhere effect across the six statistics is tracked but not globally corrected, the parity $p = 0.018$ over six trials is not globally significant. No Bianchi or source-identification claim is made. The map is full-sky cleaned with no mask deconvolution or full pixel–pixel covariance.

4.2 K4 — CF4++ bulk-flow apex versus depth

From the Cosmicflows-4 (CF4) reconstruction grid we measure the volume-averaged bulk-flow amplitude and apex direction in seven radial shells. Table 2 gives the profile.

Findings. The bulk-flow amplitude rises to ~ 590 km/s at 250–300 Mpc/h and then falls to ~ 130 –190 km/s in the deep shells. The apex stays within ~ 5 – 46° of the full-sample apex out to 300 Mpc/h and then reverses (~ 120 – 149°) at the reconstruction edge; the reversal is flagged as

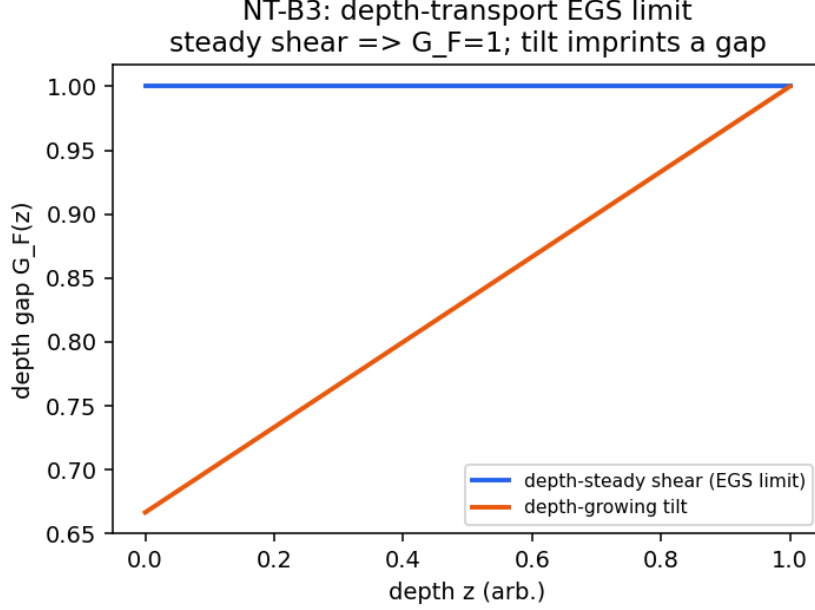


Figure 3: Theorem 3 (NT-B3): the depth gap $G_F(z)$ is flat at unity for a depth-steady shear (EGS depth limit) and rises when a depth-evolving tilt is present. Diagnostic-only; not a detection.

Table 2: K4: CF4++ bulk-flow apex by depth shell. $|B|$ is the volume-averaged reconstruction velocity per shell; the bracketed figures are bootstrap apex dispersion over correlated reconstruction cells (a lower bound on direction uncertainty, not a full covariance). Full-sample bulk flow: 133.4 km/s toward $(l, b) = (121.6^\circ, 70.8^\circ)$.

r [Mpc/h]	N cells	$ B $ [km/s]	apex (l, b) [deg]	to CMB apex [deg]
90–150	6749	378	(300, 24)	36.9
150–200	10254	531	(301, 63)	25.3
200–250	16091	584	(239, 88)	39.9
250–300	21788	588	(133, 75)	52.5
300–350	27890	222	(114, 48)	80.4
350–400	35806	128	(114, −49)	160.6
400–450	45179	186	(120, −78)	147.5

a likely edge effect of the reconstruction. This is a model-independent kinematic descriptor of the reconstruction; it is not a cosmological-frame-violation or Bianchi claim, and the bootstrap dispersion is a lower bound on the true direction uncertainty rather than a full covariance.

Critically excluded candidates. Several nearby analyses were considered and *excluded* as empty or as re-presentations, to keep this report free of non-results: a joint common-axis posterior over the four dipoles (the available latent likelihood has no directional term, so the axis is prior-flat); the DESI footprint resultant-dipole “systematics floor” (already shipped as a long-run jackknife diagnostic); the DESI cosmological clustering dipole (blocked: no matched randoms/masks); and any positive Bianchi/tilt evidence (blocked: amplitude-matched contamination null, no native morphology atlas).

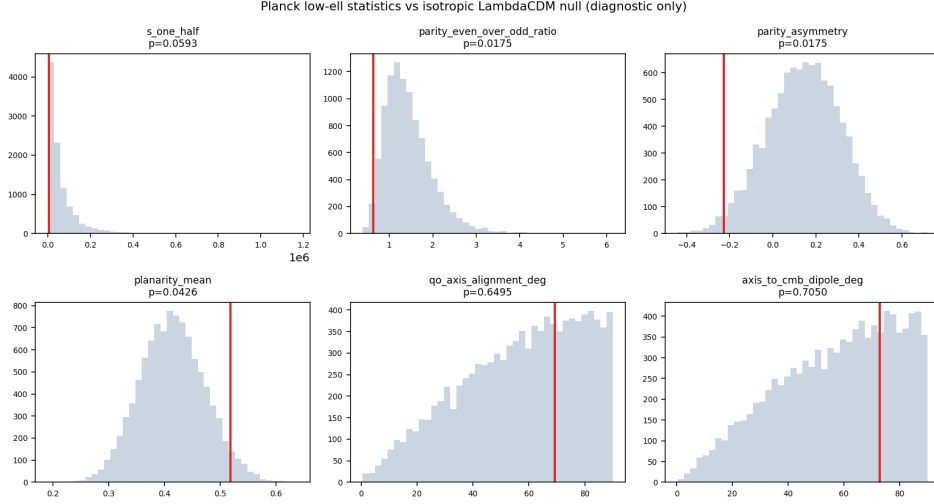


Figure 4: K1: observed low- ℓ statistics of the Planck PR3 map against the isotropic Λ CDM null ensemble. Model-independent null-calibrated features; not evidence for any Bianchi model.

5 Transfer-conditional diagnostics

A separate family of result surfaces depends on *external* or *empirical-proxy* transfer functions (AniCLASS-external and an empirical shear $\rightarrow D_\ell$ proxy). These are *transfer-conditional* diagnostics: they are not native low- ℓ solver output, they carry no passed native-validation gate, and the family labels attached to them are provenance labels only, not classification claims. Any amplitude-ranking quantity computed along these paths (for example a configured likelihood-ratio summary for an admitted dipole-amplitude premise) is a premise-conditioned amplitude fit and a transfer-conditional ranking; it is prior- and error-budget-sensitive, it is not a source-identification claim, and the source origin is not established. The auditable inventory of transfer dependence is `docs/generated/transfer_sensitivity_report.md`; the calibration window for the external shear $\rightarrow D_2$ path is the legacy Saadeh-linear regime $\Sigma^2 \in [10^{-24}, 10^{-20}]$. These rows are included for completeness and provenance, not as headline results.

6 What is not claimed (standing blocks)

For the avoidance of doubt, the following are *not* results of this report and remain blocked:

- an identified Bianchi family or a detected anisotropic geometry from any scalar, axis, or low- ℓ feature above;
- native-solver validation for any external/proxy transfer output;
- any MIO observatory certificate used as a posterior, a truth certificate, a model weight, or an HTT evidence term;
- any low- ℓ or velocity feature above promoted from a model-independent measurement into anisotropy evidence;
- a globally significant low- ℓ anomaly detection (the K1 p -values are local, look-elsewhere-tracked, single-sky, and not full-covariance/mask-coupled).

7 Reproducibility

Every result above is regenerated by a recorded command and ships a manifest with input hashes and a claim tier. The load-bearing artifacts are:

Planck PR3 SMICA low- ℓ ($\ell=2,3$) reconstruction

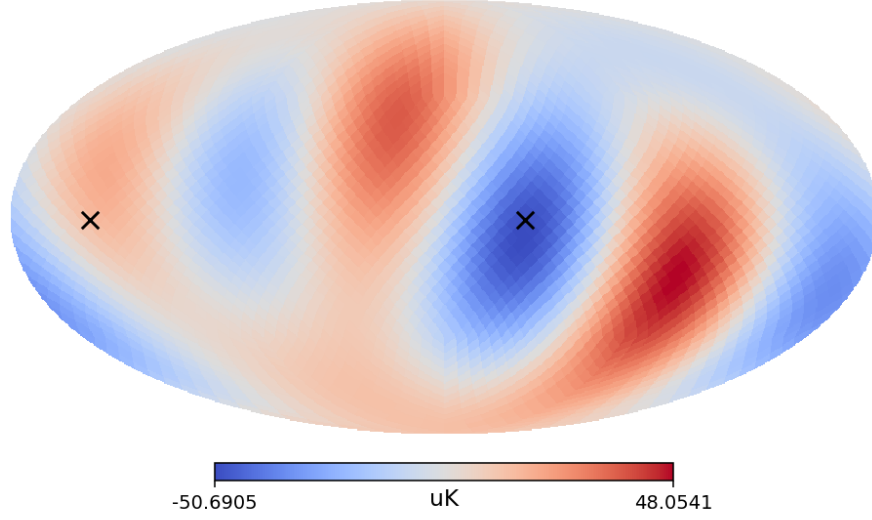


Figure 5: K1: the power-inertia low- ℓ ($\ell = 2, 3$) preferred axis of the real Planck map. The axis-alignment and axis-to-apex angles are null (Table 1). Diagnostic-only; a power-inertia-axis descriptor, not the multipole-vector axis-of-evil statistic.

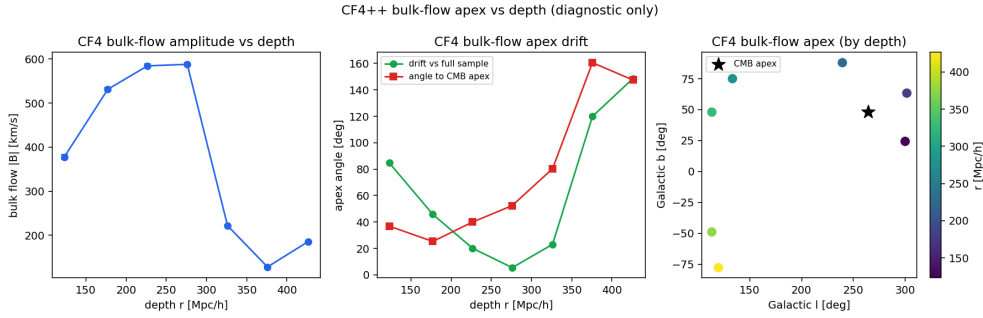


Figure 6: K4: CF4++ bulk-flow amplitude and apex drift versus depth shell. Model-independent kinematic descriptor; not a frame-violation or Bianchi claim.

- **Theorems:** `python scripts/prove_egs_lowell_theorems.py → docs/generated/egs_lowell_theorem.py` figures via `python scripts/make_egs_lowell_theorem_figures.py`.
- **K1 (Planck):** `python scripts/make_lowell_morphology_real_map.py → docs/generated/lowell_morphology.py`
- **K4 (CF4):** `python scripts/make_cf4_bulkflow_apex_depth.py → docs/generated/cf4_bulkflow_apex.py`
- **Transfer inventory:** `python scripts/generate_transfer_sensitivity_report.py`.

Each figure in this report has a sidcar `*.manifest.json` recording owner, claim tier (*diagnostic-only*), `family_identification:false`, and `native_solver_result:false`.

References

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